De Mathematics Competitions

4th Annual

DMC 10 A

Wednesday, September 30, 2026



INSTRUCTIONS

1. DO NOT TURN TO THE NEXT PAGE UNTIL YOUR COMPETITION MANAGER TELLS YOU TO BEGIN.
2. This is a 25-question multiple-choice competition. For each question, only one answer choice is correct.
3. Mark your answer to each problem on the answer sheet with a #2 pencil. Check blackened answers for accuracy and erase errors completely. Only answers that are properly marked on the answer sheet will be scored.
4. SCORING: You will receive 6 points for each correct answer, 1.5 points for each problem left unanswered, and 0 points for each incorrect answer.
5. Only blank scratch paper, rulers, compasses, and erasers are allowed as aids. No calculators, smartwatches, phones, or computing devices are allowed. No problems on the competition will require the use of a calculator.
6. Figures are not necessarily drawn to scale.
7. Before beginning the competition, your competition manager will ask you to record your name on the answer sheet.
8. You will have 75 minutes to complete the competition once your competition manager tells you to begin.
9. When you finish with the competition, sign your name in the space provided on the answer sheet and complete the demographic information questions on the back of the answer sheet.

The problems and solutions for this DMC 10 were prepared
by the DMC Committee under the direction of:

Taiki Aiba and Ryan Shin

The DMC Committee reserves the right to disqualify scores from a school if it determines that the rules or the required security procedures were not followed.

The publication, reproduction or communication of the problems or solutions of this competition during the period when students are eligible to participate seriously jeopardizes the integrity of the results. Dissemination via phone, email, or digital media of any type during this period is a violation of the competition rules.

Students who score well on this DMC 10 will be invited to take the De Invitational Mathematics Examination (DIME), which is already out and is due on Friday, January 29, 2027. More details about the DIME can be found on our site.

1. What value of n satisfies the following equation?

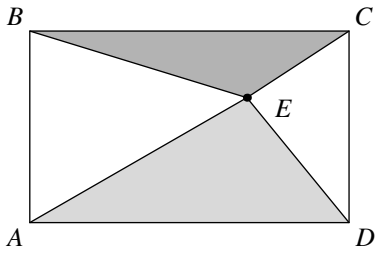
$$\frac{1}{4} + \frac{1}{6} + \frac{1}{8} = \frac{n}{192}$$

(A) 88 **(B)** 92 **(C)** 96 **(D)** 100 **(E)** 104

2. Liam naps from 2:35 to 3:23. Emma starts baking a cake at 2:55 and bakes it for one-third as long as Liam naps. At what time does the cake finish baking?

(A) 3:07 **(B)** 3:09 **(C)** 3:11 **(D)** 3:13 **(E)** 3:15

3. Point E lies inside rectangle $ABCD$, as shown below. The area of $ABCD$ is 46, and the area of $\triangle AED$ is 15. What is the area of $\triangle BEC$?



(A) $\frac{15}{2}$ **(B)** 8 **(C)** $\frac{31}{3}$ **(D)** $\frac{23}{2}$ **(E)** 15

4. At Test Academy, 50% of the students had eggs for breakfast, 60% had pancakes, and 20 students had both. Every student had at least one of the two. How many students attend Test Academy?

(A) 100 **(B)** 160 **(C)** 200 **(D)** 240 **(E)** 400

5. What fraction of the sum $1 + 2 + 3 + \cdots + 49$ comes from the terms that are multiples of 7?

(A) $\frac{1}{7}$ **(B)** $\frac{4}{25}$ **(C)** $\frac{8}{49}$ **(D)** $\frac{1}{6}$ **(E)** $\frac{1}{5}$

6. Noah has a 12-hour digital clock that skips every time whose display contains the digit 3. For example, one minute after correctly showing 12:02, it shows 12:04. Noah sets the clock to the correct time, 12:00. What is the actual time when the clock first shows 6:00?

(A) 3:15 **(B)** 3:30 **(C)** 3:35 **(D)** 3:45 **(E)** 4:00

7. A school day has 7 periods, with recess between periods 2 and 3 and lunch between periods 5 and 6. The periods are English, Math, Music, Sports, Computers, and two identical Science periods, which must be consecutive and not separated by recess or lunch. How many schedules are possible?

   (A) 240 **(B)** 360 **(C)** 480 **(D)** 600 **(E)** 720

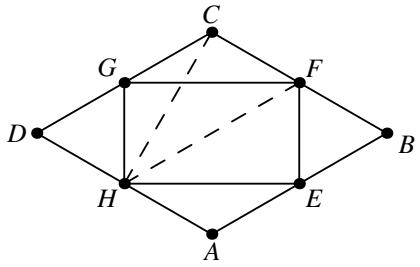
8. Twelve people, including Lucas, sit at a round table. In how many ways can 5 of the people be chosen, including Lucas, so that no two chosen people sit next to each other?

   (A) 12 **(B)** 15 **(C)** 21 **(D)** 35 **(E)** 36

9. A list of positive integers begins with 20^{26} and ends with 1. Each term after the first is a divisor of the term before it and is smaller than that term. What is the greatest possible number of terms in the list?

   (A) 27 **(B)** 53 **(C)** 78 **(D)** 79 **(E)** 80

10. In rhombus $ABCD$, $\angle ABC = 60°$. Points E, F, G, and H are the midpoints of $\overline{AB}$, $\overline{BC}$, $\overline{CD}$, and $\overline{DA}$, respectively. What is the degree measure of $\angle CHF$?



   (A) 30 **(B)** 36 **(C)** 45 **(D)** 60 **(E)** 75

11. Ethan walks his dog at least once in any given 3 consecutive days and at least 3 times in any given 7 consecutive days. What is the least number of days in May, which has 31 days, on which he can walk his dog?

   (A) 9 **(B)** 10 **(C)** 11 **(D)** 12 **(E)** 13

12. Let $P(x)$ be a quadratic polynomial with leading coefficient 1 such that $P(1) = 4$ and

$$P(x) + P\left(\frac{1}{x}\right) = P\left(x + \frac{1}{x}\right)$$

   for all nonzero real x. What is $P(5)$?

   (A) 32 **(B)** 35 **(C)** 37 **(D)** 40 **(E)** 42

13. Circles ω_1 and ω_2, each with radius 3, are internally tangent to a circle Ω with radius 9. Their points of tangency are chosen independently and uniformly at random on Ω. What is the probability that ω_1 and ω_2 intersect?

(A) $\frac{1}{6}$ **(B)** $\frac{1}{4}$ **(C)** $\frac{1}{3}$ **(D)** $\frac{1}{2}$ **(E)** $\frac{2}{3}$

14. Chloe has a 5×7 sheet of paper. Each move, Owen chooses one piece and tears it into two pieces of equal area. After some number of moves, exactly 5 pieces have area less than 1, and every other piece has area greater than 2. What is the least possible number of moves?

(A) 8 **(B)** 9 **(C)** 10 **(D)** 11 **(E)** 12

15. Isabella draws a card at random from five cards numbered 0, 1, 2, 3, 4 and stands at that point on a number line. Each second, she steps 1 unit left or right with equal probability, except that once she is at 0 or 4 (including at the start), she stays there forever. What is the probability that after 10 seconds she is standing on an even number?

(A) $\frac{4}{5}$ **(B)** $\frac{9}{10}$ **(C)** $\frac{19}{20}$ **(D)** $\frac{39}{40}$ **(E)** $\frac{79}{80}$

16. Let N be the number of nonzero coefficients in the expansion of

$$\left(x + 2x^2 + 3x^3 + \cdots + 2026x^{2026}\right)^2$$

that are even. What is the sum of the digits of N?

(A) 5 **(B)** 7 **(C)** 13 **(D)** 14 **(E)** 15

17. Each of 100 people is a soldier, a wizard, or a general, and every two people of different types fight exactly once. Generals defeat everyone of another type, wizards defeat soldiers, and soldiers defeat no one. The wizards won 600 fights and lost 1000 fights in total. How many fights took place?

(A) 2600 **(B)** 3100 **(C)** 3200 **(D)** 3300 **(E)** 4950

18. Let m and n be positive integers such that

$$\gcd(m, 20) \cdot \gcd(n, 26) + \operatorname{lcm}(m, 20) \cdot \operatorname{lcm}(n, 26) = 780,$$

where $\gcd(a, b)$ denotes the greatest common factor of positive integers a and b, while $\operatorname{lcm}(a, b)$ denotes their least common multiple. What is the least possible value of $m + n$?

(A) 30 **(B)** 33 **(C)** 36 **(D)** 39 **(E)** 42

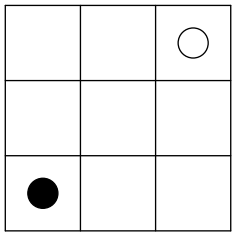
19. In $\triangle ABC$, $\angle ABC = 26^\circ$. Point D lies on $\overline{BC}$ so that $\overline{AD}$ bisects $\angle BAC$, and $AD > CD$. What is the greatest possible integer value of the degree measure of $\angle ADC$?

    (A) 52 **(B)** 68 **(C)** 77 **(D)** 78 **(E)** 102

20. While watching a show, Ayako, Billy, Carlos, Dahlia, Ehuang, and Frank sat in a row of six chairs. Ayako sat somewhere to the right of Billy, Carlos sat somewhere to the right of both Billy and Dahlia, and Ehuang sat somewhere to the right of both Dahlia and Frank. In how many orders could the six have been seated?

    (A) 32 **(B)** 45 **(C)** 52 **(D)** 61 **(E)** 90

21. In the 3×3 grid shown below, a black token and a white token are placed in the given cells. Daniel has two buttons, U and R, where pressing U moves the token one cell up, and pressing R moves it one cell right.

    However, the black token cannot move out of the grid, so if a button press results in a move that causes this, then the token moves in the opposite direction (that is, it moves left instead of right and down instead of up). How many possible sequences of 8 button presses are there in which the black token ends at the white token?

    (A) 70 **(B)** 96 **(C)** 104 **(D)** 126 **(E)** 150

22. In $\triangle ABC$ with $\angle B = 90^\circ$ and $AB = BC = 2\sqrt{2}$, circle ω passes through the midpoints of $\overline{AB}$, $\overline{BC}$, and $\overline{CA}$. Points D and E lie on ω with $\angle DBE = 90^\circ$ and $\angle BDE = 60^\circ$, and D is closer to A than E is. What is the area of non-convex pentagon $ACEBD$?

    (A) $2 + \frac{\sqrt{3}}{2}$ **(B)** 3 **(C)** $\frac{5 + \sqrt{3}}{2}$ **(D)** $2 + \sqrt{3}$ **(E)** $2 + 2\sqrt{3}$

23. Let $1, 11, 111, \ldots$ be the increasing sequence of positive integers whose digits are all 1s. For which n is the sum of the digits of the square of the nth term equal to 2026?

    (A) 136 **(B)** 179 **(C)** 181 **(D)** 226 **(E)** 2026

24. Let positive integers $a_1, a_2, a_3, \ldots$ satisfy $a_{n+2} = a_{n+1} + a_n$ for all $n \geq 1$. Suppose that one of the integers equals 2026. What is the least possible value of a_3?

    (A) 24 **(B)** 46 **(C)** 68 **(D)** 80 **(E)** 92

25. A sequence of positive integers $a_1, a_2, a_3, \ldots, a_{11}, b_1, b_2, b_3, \ldots, b_{11}$ satisfies $a_1 = b_1 = 1$ and $a_{11} = b_{11} = 27$. For each $2 \leq i \leq 11$, $a_i - a_{i-1}$ is a positive even integer, and $b_i - b_{i-1}$ is a positive odd integer. Also, $a_k = b_k$ for every odd k. How many such sequences are there?

    (A) 1440 **(B)** 10,880 **(C)** 20,160 **(D)** 21,760 **(E)** 43,520